

Computer Architecture

Problem Set #2

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This problem set is **ungraded and not collected**. We will release solutions two weeks after the problem set is released. Please stop by office hours if you have questions.

1. Draw the pipeline diagram for the execution of the following code on the two-way superscalar, in-order processor shown in Figure 1. Assume the following:

- Branches execute in **Pipeline A**.
- Loads/stores execute in **Pipeline B**.
- Full bypassing is used wherever possible.
- There are no alignment problems.

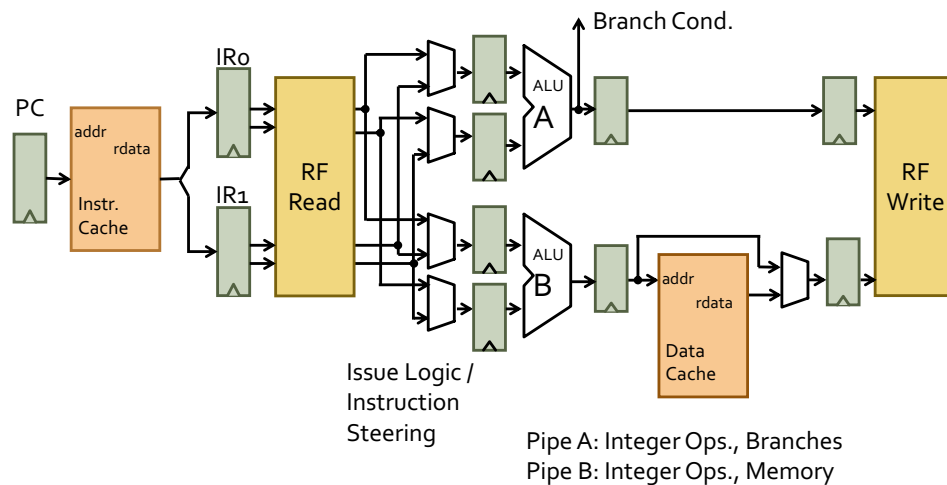


Figure 1: Figure of Two-Wide In-order Superscalar Processor Pipeline

```
add    x5, x6, x7
sub    x6, x7, x8
lw     x10, 0(x6)
addi   x12, x13, 1
lw     x15, 4(x6)
lw     x15, 4(x15)
add    x6, x9, x10
addi   x8, x10, x11
```

2. Consider an architecture with the following pipeline stages:

Assume that register fetch occurs in the **I** stage, and branch resolution takes place in the **X1** stage.

1. If a branch is mispredicted and taken, how many instructions need to be flushed (killed) from the pipeline?
2. Now, assume the pipeline is a three-wide superscalar. How many instructions must be flushed on a branch misprediction?

(Assume that the branch is the first instruction executing after a jump)

3. A processor is executing a program when it encounters a divide instruction that attempts to divide by zero, triggering an exception. Answer the following questions about how a generic processor handles such events:
 - (a) When an exception such as division by zero occurs, what information must the processor save so that program execution can resume correctly after the handler finishes?
 - (b) Suppose the exception handler reads registers R5, R6, and R7, and writes to registers R5, R8, and R10. Which registers must the handler save to preserve the program's state?
 - (c) When the handler returns control to the program, what changes does the "return-from-interrupt" instruction make to the processor's state?
4. An instruction encounters the following synchronous exceptions: an Instruction Address Exception and an ALU Overflow.

What value should be loaded into the interrupt cause register? Additionally, what happens if there is an external interrupt pending for the same instruction? Provide an explanation.

5. Assume you have the IO2I pipeline from the lecture. This pipeline can issue one instruction per cycle and commit one instruction per cycle. Draw the pipeline diagram for the execution of the following code sequence:

```
mul  x6,  x7,  x8
add  x9,  x10, x11
add  x11, x12, x13
add  x13, x14, x15
add  x19, x13, x10
lw   x2,  0(x3)
add  x12, x16, x19
lw   x5,  0(x2)
add  x15, x20, x21
```

6. Assume the I2OI architecture from the lecture. Draw the state of the scoreboard when instruction 3 is in the "I" (Issue) stage of the pipeline for the following code sequence:

```
0: mul  x6,  x7,  x8
1: add  x9,  x6,  x11
2: mul  x7,  x1,  x2
3: lw   x10, 0(x1)
```

7. The following code is to be executed on a processor with 32 architectural registers that can issue instructions out-of-order. The processor is a single-issue machine, and the functional units have varying latencies: multiplication operations take 4 cycles, ALU operations take 1 cycle, and load/store operations take 2 cycles. The processor stalls on WAW (Write After Write) and WAR (Write After Read) dependencies. As a compiler, make changes to the following code to improve performance on this out-of-order processor. Assume that all registers not currently used are available for use. You may either modify the code directly or provide a rewritten version as necessary.

[Code sequence]

```
0:  mul      x5, x6, x7
1:  add      x8, x5, x6
2:  mul      x10, x13, x8
3:  sw       x12, 0(x10)
4:  sub      x10, x6, x4
5:  mul      x17, x3, x10
6:  addi     x3, x5, 1
```